

Mobile and Wireless Security

Colton Schmid

Ferris State University

ISIN 121

Dr. Jasun Hawkins

11/5/2024

Mobile and Wireless Security

When it comes to network security there are many ways to not only make sure that your data is secure within your network but also to make sure that the network itself is secure.

Whether it is a public network or your own personal network, there are certain things you should do to ensure the protection of your network and data.

Public Wi-Fi Networks

Public networks include hotspots, airport Wi-Fi, and other places that offer free Wi-Fi to the public. These networks in the early days of the Internet were extremely unsafe and anyone could access your information if they knew how to look for it. However, as the internet has grown these networks have become more secure by encrypting the network “to scramble the data and protect it from hackers snooping on the network” (Federal Trade Commission, 2022).

Although some networks do this, they are not required, so you should check to see if there is a lock or “https” in the address bar. This ensures that your data is safe from attackers when using that website.

Utilizing the best security practices while on a public network and be vital to protecting your information from attackers. One of these is a strong password and turning on two-factor authentication. This not only adds another layer of security but it can notify you of a password breach. Another good security practice is just being aware of what attacks look like, such as phishing or typo-squatting, and avoiding scammers.

Securing Your Wireless Network

Now when it comes to your home or personal network there are some similar things that you should do to ensure the security of your data and network, but there are also multiple different steps you should take. The first step you should take is to encrypt your network. This

makes the data on the network unrecognizable to anyone trying to view or access it without the encryption key. One way that you can do this is by “updat[ing] your router settings to either WPA3 Personal or WPA2 Personal” (Federal Trade Commission, 2021). Another step you should take to increase your security is changing your router's default settings. Changing the default settings for the admin username and password, or the network name, to something unique can make it more difficult for attackers to successfully breach your data. The network password and the admin password should 100% be changed to something unique because if these are breached they can gain access to the majority of your data. Keeping your router up to date can also greatly increase your security. This is because if there is an update it most likely repairs holes in the system, easily increasing your security. Adding a guest network can also be a lifesaver because if someone outside of your home joins your network and unknowingly has malware on their device, your whole network could be compromised. Last but not least, having a firewall active can prevent many attacks. This is because a firewall monitors the traffic on the network and detects if something is malware or should not be there. In turn, this blocks it from entering the network, saving you from a data breach or attack.

wireless security practices

There are many ways to practice good wireless security habits, and many professionals think that the same actions can greatly increase your security on the internet. The main suggestion is to have a strong and unique password, while also enabling two-factor authentication as these will greatly decrease your risk of an attack or breach to occur. Another strongly recommended action is encrypting your data. This is because encrypting your data makes it nearly unrecognizable to anyone who is not supposed to access it. Additionally, using a VPN and installing a firewall, both personal and network, Can greatly increase your security and

protection from attacks and breaches. In addition to these practices, there are many ways that you can increase your security, I think one of the biggest ways is just being aware of what attacks and breaches can look like. This is because you can avoid many situations that would put you there by knowing what they look like.

Education. When it comes to education there are many ways to educate yourself or even friends and family to, hopefully, prevent them from a future attack. The first way that I would educate someone is by telling them a detailed description and examples of each known type of attack currently in cyberspace. To explain, I would show them an example of a phishing email or a social engineering example then I would explain them in detail. I think this is a great way to educate someone because knowing what an attack looks like, how to prevent it, and how to deal with it can be vital to someone who uses the internet. After that, I would educate them on preventing and decreasing their risks of attacks.

References

Baig, A. (2022, June 17). 12 Best Practices for Wireless Network Security. GlobalSign.

<https://www.globalsign.com/en/blog/12-best-practices-wireless-network-security>

CISA. (2021, February 1). Securing Wireless Networks. Cybersecurity and Infrastructure Security Agency CISA.

<https://www.cisa.gov/news-events/news/securing-wireless-networks>

Federal Trade Commission. (2022, February 9). Are Public Wi-Fi Networks Safe? What You Need To Know. Consumer Advice.

<https://consumer.ftc.gov/articles/are-public-wi-fi-networks-safe-what-you-need-know>

FTC. (2018, October 5). Cybersecurity Basics. Federal Trade Commission.

<https://www.ftc.gov/business-guidance/small-businesses/cybersecurity/basics>

Federal Trade Commission. (2021, May 27). How To Secure Your Home Wi-Fi Network. Consumer Information.

<https://consumer.ftc.gov/articles/how-secure-your-home-wi-fi-network>

